Hints: Polynomials – 6. Polynomials over finite fields

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. At most n roots: if $f(a) = 0$ then $f = (x-a)g$ with $\deg g = n-1$, and a root $b \neq a$ of f is a root of g (a field has no zero divisors); induct. Every $a \in \mathbb{F}_q^*$ satisfies $a^{q-1} = 1$ (Lagrange in the group $\mathbb{F}_q^*$), so all q elements are roots of the monic degree-q polynomial $x^q - x$, which forces $x^q - x = \prod_a (x-a)$. A function is given by Lagrange interpolation, and two polynomials of degree $< q$ agreeing at q points coincide.

Hint 2. Cyclicity: let $\psi(d)$ be the number of elements of order d ($d \mid q-1$). If $\psi(d) > 0$ and g has order d, then $1, g, \ldots, g^{d-1}$ are all the roots of $x^d - 1$, so the elements of order d are among these powers and $\psi(d) = \varphi(d)$. Hence $\psi(d) \leqslant \varphi(d)$, and $\sum_{d \mid q-1} \psi(d) = q - 1 = \sum_{d \mid q-1} \varphi(d)$ gives $\psi(q-1) = \varphi(q-1) > 0$.

Hint 3. Wilson: compare constant terms in $x^{p-1} - 1 = \prod_{a \neq 0}(x-a)$ over $\mathbb{F}_p$. k-th powers: with a generator g and $d = \gcd(k, q-1)$, g^m is a k-th power iff $d \mid m$, iff $a^{(q-1)/d} = 1$.

Problem 2

Hint 1. A polynomial of degree 2 or 3 that is reducible has a factor of degree 1, i.e. a root.

Hint 2. For $n \geqslant 4$ build a reducible polynomial of degree n with no root: multiply an irreducible quadratic by an irreducible polynomial of degree $n-2$ (over a finite field irreducible polynomials of every degree exist, e.g. by the counting formula of Theorem 2). Answer: $n = 2, 3$.

Problem 3

(a)

Hint 1. $x \equiv y \pmod{n}$ implies $x - y \mid f(x) - f(y)$, so $f(x) \equiv f(y) \pmod{n}$.

(b)

Hint 1. A surjective map of a finite set to itself is a bijection, so f_n is injective for every $n > 2$. Also $a - b \mid f(a) - f(b)$ for all integers a, b.

Hint 2. $f(a) \neq f(b)$ for $a \neq b$ (take $n > \max(2, |a-b|)$). If $m = |f(a) - f(b)| > 2$, injectivity of f_m gives $m \mid a - b$; with $a - b \mid f(a) - f(b)$ this means $|f(a) - f(b)| = |a - b|$. Deduce that this equality holds whenever $|a - b| \geqslant 3$.

Hint 3. So $f(x) - f(0) = \pm x$ for infinitely many x, hence $f(x) = \pm x + c$ identically; these clearly work. Answer: $f(x) = x + c$ or $f(x) = -x + c$, $c \in \mathbb{Z}$.

Problem 4

Hint 1. n is interesting iff $x^p - x + 1$ divides $x^n - 1$ in $\mathbb{F}_p[x]$, i.e. $x^n = 1$ in the ring $R = \mathbb{F}_p[x]/(x^p - x + 1)$. Use the Frobenius in R.

(a)

Hint 1. In R, $x^p = x - 1$, and since $u \mapsto u^p$ is a ring homomorphism fixing $\mathbb{F}_p$, $x^{p^k} = x - k$. So $x^{p^p} = x - p = x$, and x is invertible in R (the constant term of the modulus is 1), giving $x^{p^p - 1} = 1$.

(b)

Hint 1. The interesting numbers are the multiples of the order of x in R^*, so the question is when this order is $p^p - 1$. The roots of $x^p - x + 1$ are $\theta, \theta - 1, \ldots, \theta - (p-1)$, permuted cyclically by Frobenius, so the polynomial is irreducible and $R = \mathbb{F}_{p^p}$.

Hint 2. Compute the norm: $x^{1+p+\cdots+p^{p-1}} = x(x-1)\cdots(x-p+1)$ in R, which is the product of the roots, equal to -1 for odd p. Hence $x^{2(p^p-1)/(p-1)} = 1$, which is a smaller exponent when $p \geqslant 5$.

Hint 3. For $p = 2$: $x^2 + x + 1$, order $3 = 2^2 - 1$. For $p = 3$: the order divides 26; $x^{13} = -1$ by the norm and $x^2 \neq 1$. Answer: $p = 2$ and $p = 3$.

Problem 5

Hint 1. Work over $\mathbb{F}_p$. When does the determinant $D(x, y, z)$ vanish? Use that $u \mapsto u^p$ is additive and fixes $\mathbb{F}_p$.

Hint 2. If $c \neq 0$ and you substitute $z = -(ax+by)/c$ with $a, b, c \in \mathbb{F}_p$, the third column becomes a combination of the first two, so $D = 0$. Hence $ax + by + cz$ divides D in $\mathbb{F}_p[x, y, z]$ for every nonzero $(a, b, c) \in \mathbb{F}_p^3$.

Hint 3. Take one form per line through the origin: $p^2 + p + 1$ pairwise non-proportional linear forms, all dividing D (unique factorization), and $\deg D = 1 + p + p^2$. So $D = \lambda \prod(ax + by + cz)$ with $\lambda \in \mathbb{F}_p^*$ (the monomial $xy^p z^{p^2}$ shows $D \neq 0$); absorb λ into one factor and lift to integers.